

Time Series Forecast of April 2025 CO₂ Levels at The Mauna Loa Observatory

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1 Introduction

Carbon dioxide (CO₂) concentration in Earth's atmosphere is one of the most closely monitored indicators of global climate change. Among the most reliable sources of atmospheric CO₂ data is the Mauna Loa Observatory in Hawaii. Located far from urban centers and above the inversion layer, the observatory provides a pristine, stable environment for measuring background CO₂ levels, free from most local emissions. The resulting dataset, initiated by Charles David Keeling in the late 1950s, forms the iconic "Keeling Curve," a central piece of evidence in studying human-driven climate change.

In this project, we aim to forecast the average CO₂ concentration at Mauna Loa for April 2025 using time series modeling techniques. Although Mauna Loa is more than 5,000 miles from Chapel Hill, the global nature of atmospheric CO₂ makes it a scientifically valuable target for statistical prediction. Our dataset includes monthly average CO₂ levels from March 2015 to February 2025—ten years of consistent observations. The data captures both long-term trends and recurring seasonal fluctuations. We approach the forecasting task by fitting several time series models to the observed data: specifically, Moving Average (MA), Autoregressive (AR), and Autoregressive Moving Average (ARMA) models. Each model is assessed using the corrected Akaike Information Criterion (AICc), which helps balance the model fit with complexity, which is especially important given the dataset's relatively short length. The model

with the lowest AICc is then used to generate a point forecast for April 2025, along with 95% and 99% prediction intervals. This report outlines our analysis, modeling process, forecast results, and the rationale behind our approach.

2 Preliminary description of the data

The dataset used in this study consists of monthly average atmospheric CO₂ levels, measured in parts per million (ppm), recorded at the Mauna Loa Observatory from March 2015 to February 2025. It includes 120 time series observations and was sourced from NOAA's official CO₂ monitoring site. A time series plot of the data shows a clear upward, linear trend in CO₂ levels over the ten years, rising from approximately 400 ppm to above 425 ppm, demonstrated in **Figure 1**. In addition to this steady increase, a pronounced seasonal pattern is evident, with consistent peaks and troughs repeating roughly every 12 months. There are no obvious structural breaks or outliers in the series, suggesting a relatively stable data-generating process apart from the trend and seasonality.

To better understand the underlying trends and seasonality components in the data, we examined the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots. The ACF plot shown in **Figure 2** demonstrates a very slow decay across many lags, with recurring peaks around lag 12 and its multiples. This behavior is typical of data with a strong trend and a seasonal component, indicating that the observations are highly correlated over time and reinforcing the conclusion that the series is non-stationary. The PACF plot shown in **Figure 3** demonstrates a large spike at lag 1, followed by smaller and mostly insignificant values at higher lags. This is characteristic of a potential autoregressive process, possibly AR(1), although this can only be confirmed after transforming the series to stationarity. Together, the ACF and PACF plots suggest that the underlying trend and seasonality displayed by the data must be addressed before applying the appropriate time series modeling techniques.

3 Heuristics of our approach

Given the strong trend and seasonal patterns observed in the CO₂ data, our modeling approach focused on capturing these features while balancing complexity and interpretability. We began by acknowledging the non-stationarity present in the original time series, as evidenced by the clear upward trend and slowly decaying autocorrelations in the ACF plot. These characteristics violate the assumptions required for many time series models, most of which typically require a stationary input. As a result, we applied different transformations to the series to make it more suitable for common time series modeling methods. After transforming the data to achieve approximate stationarity, we explored three classes of models: Moving Average (MA), Autoregressive (AR), and Autoregressive Moving Average (ARMA). MA models are useful when past error terms influence the present value, while AR models rely on past observations. ARMA combines both, allowing us to capture more complex dependencies in the data.

To guide model selection, we used the corrected Akaike Information Criterion (AICc). This metric rewards goodness-of-fit while penalizing model complexity, and it is especially useful when working with relatively short time series, like our 120-point dataset. We fit a range of candidate models within each class and selected the one with the lowest AICc. The resulting best model was then used to produce a forecast for the CO₂ level in April 2025, along with corresponding 95% and 99% prediction intervals. By comparing these three modeling classes and applying a consistent selection criterion, we ensured that our final forecast was both statistically justified and scientifically reasonable.

4 Descriptions and assumptions of our models

To model the short-term fluctuations in atmospheric CO₂ concentration, we first applied a MA model to the residual component of the time series. Before fitting the model, we converted our raw data into a time series object beginning in March 2015 with a frequency of 12

observations per year. Next, we used Seasonal-Trend decomposition using Loess (STL) to separate the original monthly CO₂ data into three parts: a long-term trend component, a repeating seasonal pattern, and a remaining residual component. This helps simplify the modeling process by allowing each component to be handled separately. The MA model focuses specifically on the residual component, which is assumed to be stationary—meaning its statistical properties, like mean and variance, remain constant over time. We selected a MA(4) model, meaning the model uses the previous four error terms to predict the current value. This choice was informed by the Partial Autocorrelation Function (PACF) plot, where a clear cut-off after lag 4 in the PACF indicated that a model including up to four lags would be appropriate. To validate the use of the Moving Average (MA) model on the STL residuals, we tested the key assumptions required for model reliability. We first confirmed stationarity using the Augmented Dickey-Fuller (ADF) test, which yielded a p-value less than 0.01. The normality of residuals was evaluated with the Shapiro-Wilk test, which returned a low p-value, suggesting some deviation from normality; however, visual inspection via Q-Q plot showed residuals were approximately symmetric, supporting the model's adequacy. To ensure uncorrelated residuals, we applied the Ljung-Box test and reviewed the autocorrelation function (ACF), both indicating that no significant autocorrelation remained. Lastly, invertibility was assumed based on the successful convergence of the *arima* function, which enforces invertibility constraints when estimating parameters. These checks support the appropriateness of our MA(4) model for short-term forecasting.

As with the MA model, we first applied STL to decompose the time series into trend, seasonal, and residual components before fitting the AR model. The residual component, assumed to be stationary, was modeled using an AR approach. Based on the PACF, we selected an AR(1) model and estimated its parameters using the *arima* function in R. The estimated AR(1) coefficient was 0.2962, with an AIC of 44.52. This model was then used to forecast the residuals two steps ahead, and both 95% and 99% confidence intervals were computed for the

predictions. For the trend component, we compared predictions from Holt's exponential smoothing and an automatically selected ARIMA model, choosing the model with the lower AIC value. The ARIMA model was ultimately selected, with an AIC of -740.6443. The seasonal component for future time points was extracted based on the existing seasonal pattern. We reconstructed the overall forecast by combining the predicted trend, seasonal, and residual components.

To properly fit ARMA models for prediction, we first had to remove the trend and seasonal components from our starting time series. Let X_t represent this original time series. By simply examining a plot of the data, we know that this series can be written as such:

$$X_t = m_t + s_t + r_t$$

In this case, m_t is the trend component, s_t is the seasonal component, and r_t is the remaining, stochastic component. We implemented Seasonal-Trend decomposition using Loess (STL), which works by iteratively smoothing and detrending the time series. First, the seasonal component is isolated by subtracting a rough estimate of the trend using Loess (Locally Estimated Scatterplot Smoothing). Loess employs weighted least squares over a moving window, assigning more weight to points near the target time. Then, for each season in the determined period (for this series, that is 12), a local regression is fitted across years to estimate values of s_t for each seasonal position. After that, the seasonal component is subtracted to focus on trend instead. By fitting Loess to $X_t - s_t$ the trend component is smoothed, and the remainder is subsequently updated. Typically, this process repeats for a pre-specified number of iterations.

With a stationary remainder, it becomes possible to fit ARMA models for prediction. After calculating the natural logarithm of the sample size (120), it is advisable to estimate no more than four or five parameters if we wish to create a stable, generalizable model. Therefore,

we limit parameters p and q , the order of the AR and MA processes, respectively, to take values less than or equal to two. Utilizing the R function *arima*, we fit all possible ARMA models on the stationary remainder under said limitation: ARMA(1, 1), ARMA(1, 2), ARMA(2, 1), and ARMA(2, 2). Each was evaluated on the corrected Akaike's Information Criterion (AICc), an estimator of prediction error and thereby quality of statistical models. Lower values of AICc indicate a better-fitting model relative to others being considered. Of all four ARMA models fit, the minimum AICc value (28.37) was associated with the ARMA model with parameters ($p = 2$, $q = 1$).

5 Estimation and inference for our models

The MA model for our analysis was chosen based on the residuals obtained from the STL decomposition of the time series. The PACF suggested that the model should include past residuals up to lag 4, leading to the selection of an MA(4) model. This model captures short-term dependencies in the residuals. The MA(4) model includes five terms: the four lagged residuals (θ_1 , θ_2 , θ_3 , and θ_4) and the intercept (μ). The coefficients represent the impact of the previous residuals (errors) on the current value, with the intercept accounting for any systematic bias in the residuals. The estimated variance of the residuals was 0.06917, and the AIC for the model was 33.94. We ran multiple diagnostic checks, including the ACF and a Ljung-Box test, which confirmed that the residuals were uncorrelated, with a p-value of 0.663 indicating no significant autocorrelation. The fitted model reflects the impact of the previous four residuals on the current value, as described by the following equation:

$$\hat{X}_t = -0.0011 + Z_t + 0.1325 * Z_{t-1} - 0.3324 * Z_{t-2} - 0.4878 * Z_{t-3} - 0.1751 * Z_{t-4},$$

where Z_t represents the error terms of the residuals.

The AR model of best fit was determined by analyzing the PACF graph of the residuals from the STL decomposition of the original time series. The PACF suggested an autoregressive model of order $p = 1$, indicating a model of best fit that includes one AR coefficient: ϕ_1 . The *arima* R function was then used to fit the AR(1) model to the residuals using maximum likelihood estimation. After estimation, the fitted AR(1) model was found to have parameter $\phi_1 = 0.2962$. The model's estimated variance was 0.082, with an AIC of 44.52. The resulting AR(1) model can be written as:

$$X_t = 0.2962 * X_{t-1}$$

The ARMA model of best fit was determined by fitting models of all possible combinations of parameter values (p, q) given a calculated limitation related to sample size and overfitting, then evaluating the fit of each model using AICc. We found the optimal parameters to be $p = 2$ and $q = 1$, meaning our model of best fit will include three estimated coefficients: Φ_1 , Φ_2 , and θ_1 . The *arima* R function estimates coefficients using maximum likelihood estimation (MLE). Given parameters p and q , the log-likelihood function of the observed data is constructed. Mathematical optimization methods are used to find coefficients that maximize the likelihood function. After estimation, the fitted ARMA(2, 1) model can be written as such:

$$X_t = 1.1584 * X_{t-1} - 0.4258 * X_{t-2} + Z_t - Z_{t-1} - 0.0011$$

It is necessary to run diagnostic checks on the ARMA(2, 1) model residuals to ensure no significant autocorrelation. A Ljung-Box test ($h = 20$) on the model residuals resulted in a p-value of 0.8205, indicating no evidence to reject the null hypothesis that the residuals are independently and identically distributed. **Figure 4** shows side-by-side ACF and PACF plots of model residuals.

6 Forecast of the monthly average of CO2 from our best model

Of all models fitted to the data, the ARMA(2, 1) model had the minimum associated AICc value of 28.37, so we use this model to forecast the monthly average carbon dioxide level for April 2025. Using the built-in R function *predict*, we estimate the monthly average carbon dioxide level two time-steps (months) ahead, according to the ARMA(2, 1) model. The function output is not equivalent to the model's actual prediction, as it is necessary to reincorporate the previously removed trend and seasonal components to obtain the real-world value.

Since we used STL decomposition before fitting the ARMA(2, 1) model, it is relatively simple to create and factor in trend and seasonal forecasts to the residual prediction. The *auto.arima* R function fits a model to the isolated trend component, and this model is then used in conjunction with the *forecast* R package to project the trend component two time-steps ahead. The month-specific seasonal value for April is easily obtained from the already isolated seasonal component.

Lastly, the trend and seasonal forecasts are added to the residual prediction, resulting in the model's final prediction of **429.65**. In context, the ARMA(2, 1) model predicts the average carbon dioxide level for April 2025 at the Mauna Loa Observatory in Hawai'i to be approximately 429.65 parts per million (ppm). The 95-percent prediction interval was calculated using the residual prediction and its standard error and can be written as **[429.14, 430.16]**. The 99-percent prediction interval can be written as **[428.98, 430.33]**.

7 Conclusion

In this project, we set out to forecast the atmospheric carbon dioxide (CO₂) concentration at the Mauna Loa Observatory for April 2025 using time series analysis. Our dataset comprised ten years of monthly CO₂ measurements, spanning from March 2015 to February 2025. Visual inspection and autocorrelation diagnostics revealed both a strong linear trend and a consistent seasonal pattern, indicating the need for decomposition before effective modeling could take

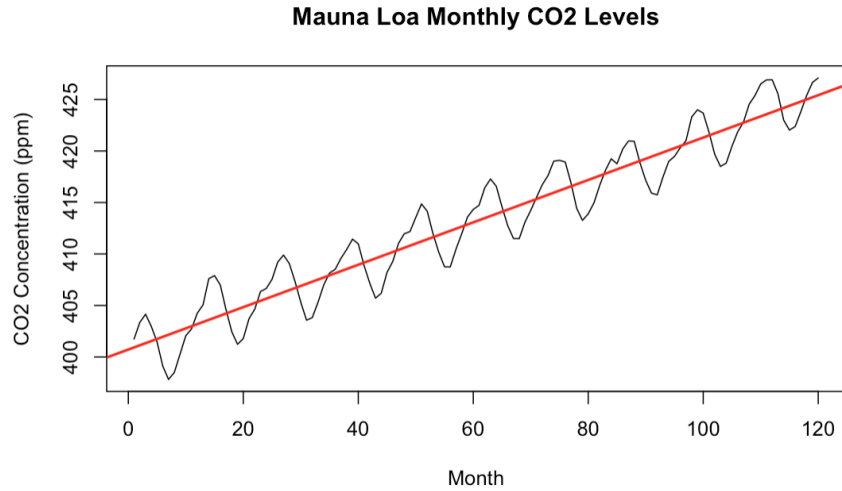
place. To address this non-stationarity, we applied Seasonal-Trend decomposition using Loess (STL), isolating the trend, seasonal, and residual components. We then fit various models to the stationary residual component, including MA(4), AR(1), and several ARMA(p, q) models. Model selection was performed using the corrected Akaike Information Criterion (AICc) to balance fit and parsimony, especially given our limited sample size of 120 months.

After a thorough comparison, the ARMA(2,1) model emerged as the best-performing candidate. It not only had the lowest AICc (28.37) among all considered models but also passed all key diagnostic checks, including the Augmented Dickey-Fuller test for stationarity and the Ljung-Box test for uncorrelated residuals. The model was used to forecast the residual component, while future trend and seasonality were estimated separately and added back to form the final forecast. Our final prediction for the average CO₂ concentration in April 2025 is 429.65 ppm, with a 95% prediction interval of [429.14, 430.16] and a 99% interval of [428.98, 430.33]. These results not only reflect the continued increase in atmospheric CO₂ levels but also validate the use of decomposition-based modeling techniques for environmental time series.

In conclusion, our analysis demonstrates the importance of addressing trend and seasonality before applying traditional time series models. By combining STL decomposition with careful model selection and validation, we produced a statistically sound and environmentally meaningful forecast. These techniques can be extended to other climate indicators or short-run policy modeling where precision and interpretability are both critical.

8 Appendix

Figure 1: Time Series for Monthly CO2 Levels at Mauna Loa from 2015 to 2025



Figures 2 and 3: ACF and PACF Plots for Mauna Loa Monthly CO2 Time Series

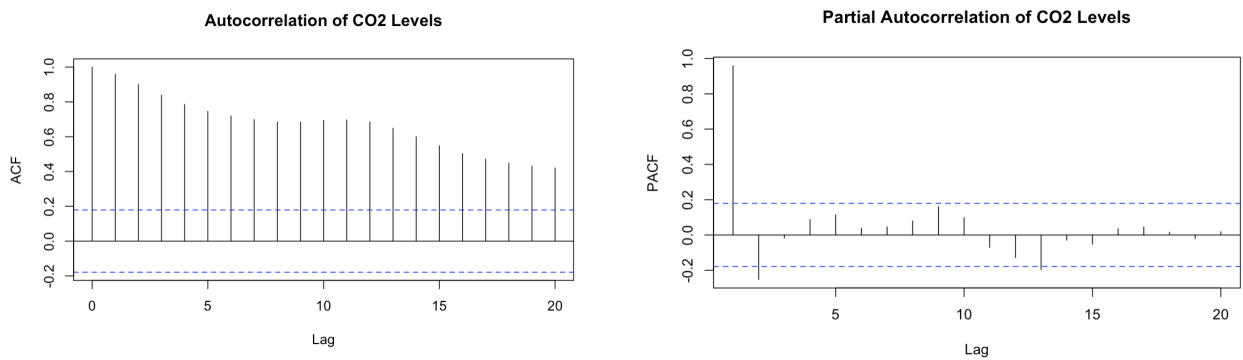


Figure 4: ACF and PACF Plots for ARMA(2, 1) Model Residuals

